

Ethics of Extended Reality Research with Children, Adult Stakeholder Perspectives

Rayne Aurit¹, Juan Pablo Hourcade²

¹University of Nebraska-Lincoln, ²University of Iowa Department of Computer Science

Introduction

What is XR?

- Virtual Reality (VR), Augmented Reality (AR)
- Ability to overlay graphic displays over reality or to completely immerse the user in a virtual world

XR and Ethics

Though introduced to older audiences, extended reality (XR) technologies are becoming adapted to suit younger audiences. Despite that most headsets produced and sold in present day are not recommended for younger children⁶, educational, therapeutic, and accessibility applications have attracted sustained research attention and government funding⁵. Both unique potential benefits and risks arise from XR technology with ethical considerations about a more vulnerable user. The project included taking a web survey to see the perspective on XR Research of stakeholders in children's lives.

Respondent Demographics*

Variable	Category	n	%
Age	18-29	66	3.3
	30-49	1369	68.5
	50-64	451	22.6
	65+	112	5.6
Sex	Female	1377	68.9
	Male	610	30.5
	Declined	13	0.7
Residence	Metropolitan	1765	88.3
	Non-metropolitan	233	11.7
	Declined	2	0.1
Education	HS or less	254	12.7
	Some college	89	4.5
	Bachelor's+	1106	55.3
	Not collected	551	27.6

Analysis

Scale Items

We used the one-sample Wilcoxon⁷ signed-rank test against the scale's empirical midpoint suited to data whose intervals cannot be assumed equal.

Ranking Items

We used the Friedman⁴ test where significant results were followed by pairwise Wilcoxon signed-rank tests between every item pair, identifying which items were preferred to which. Some questions required respondents to sort responses into ordinal categories then rank, requiring two separate tests to capture each task.

Adjustments

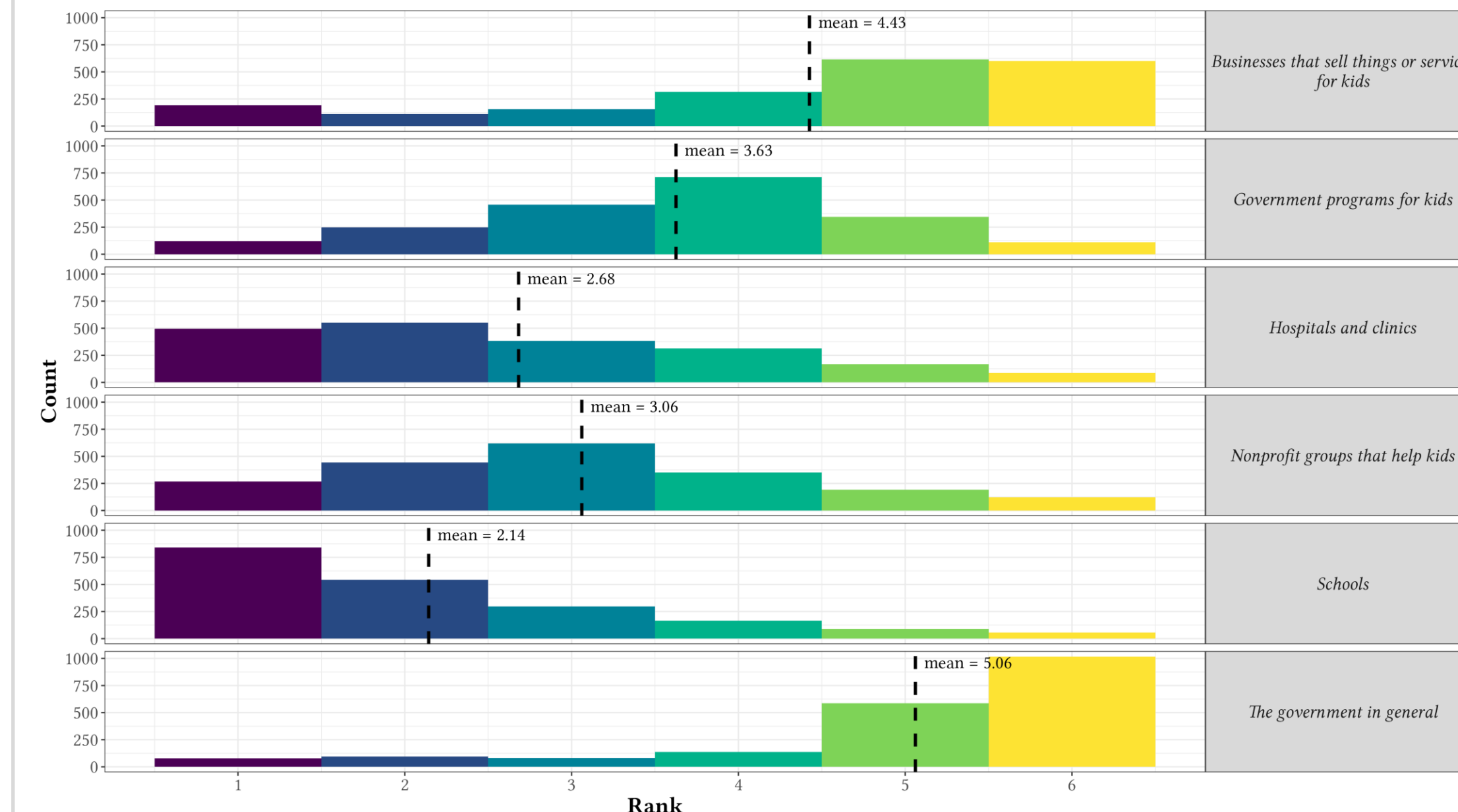
Due to many pairwise tests, p-values were corrected with the Benjamin-Hochberg adjustment³.

Results*

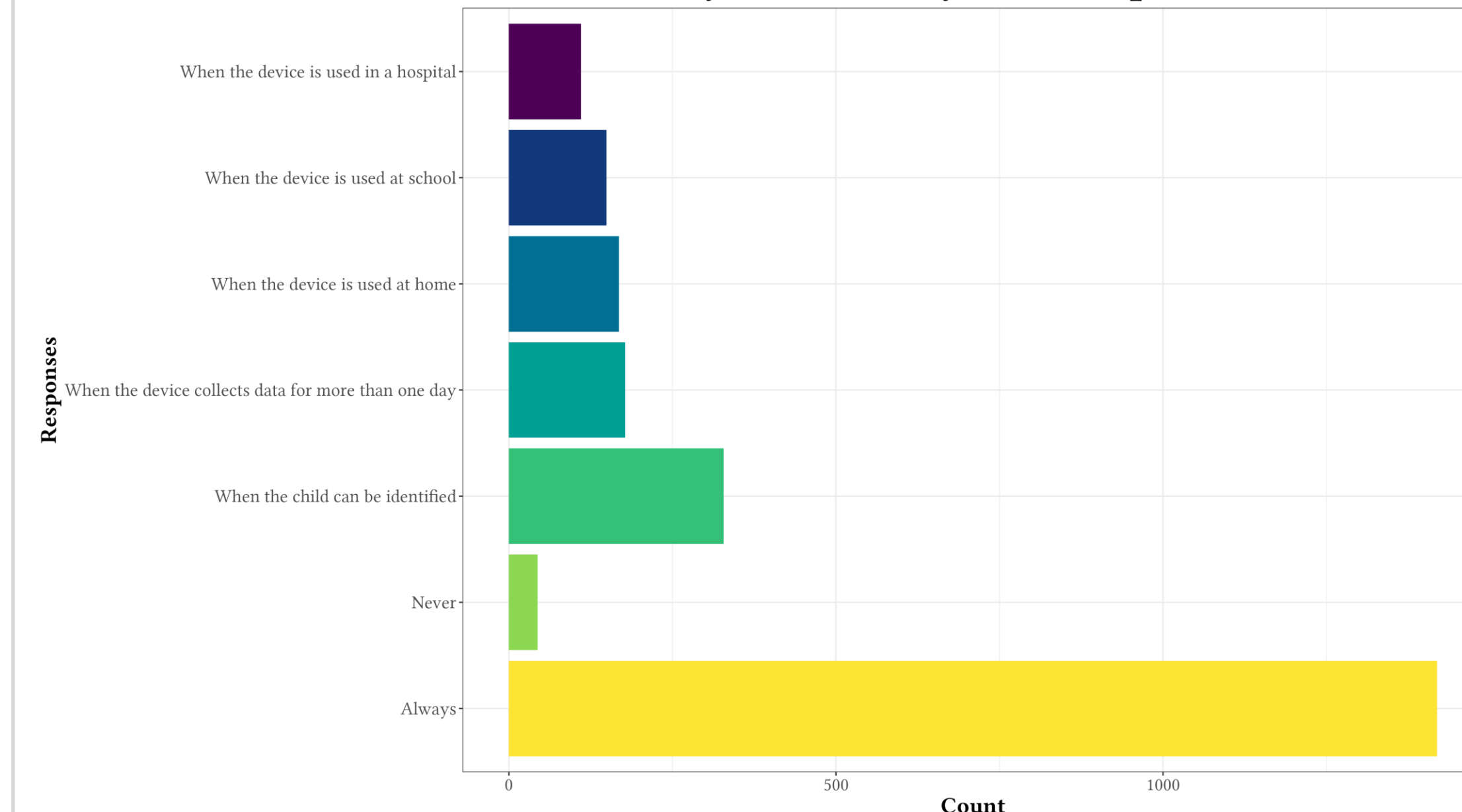
Health-related topics ranked by preference



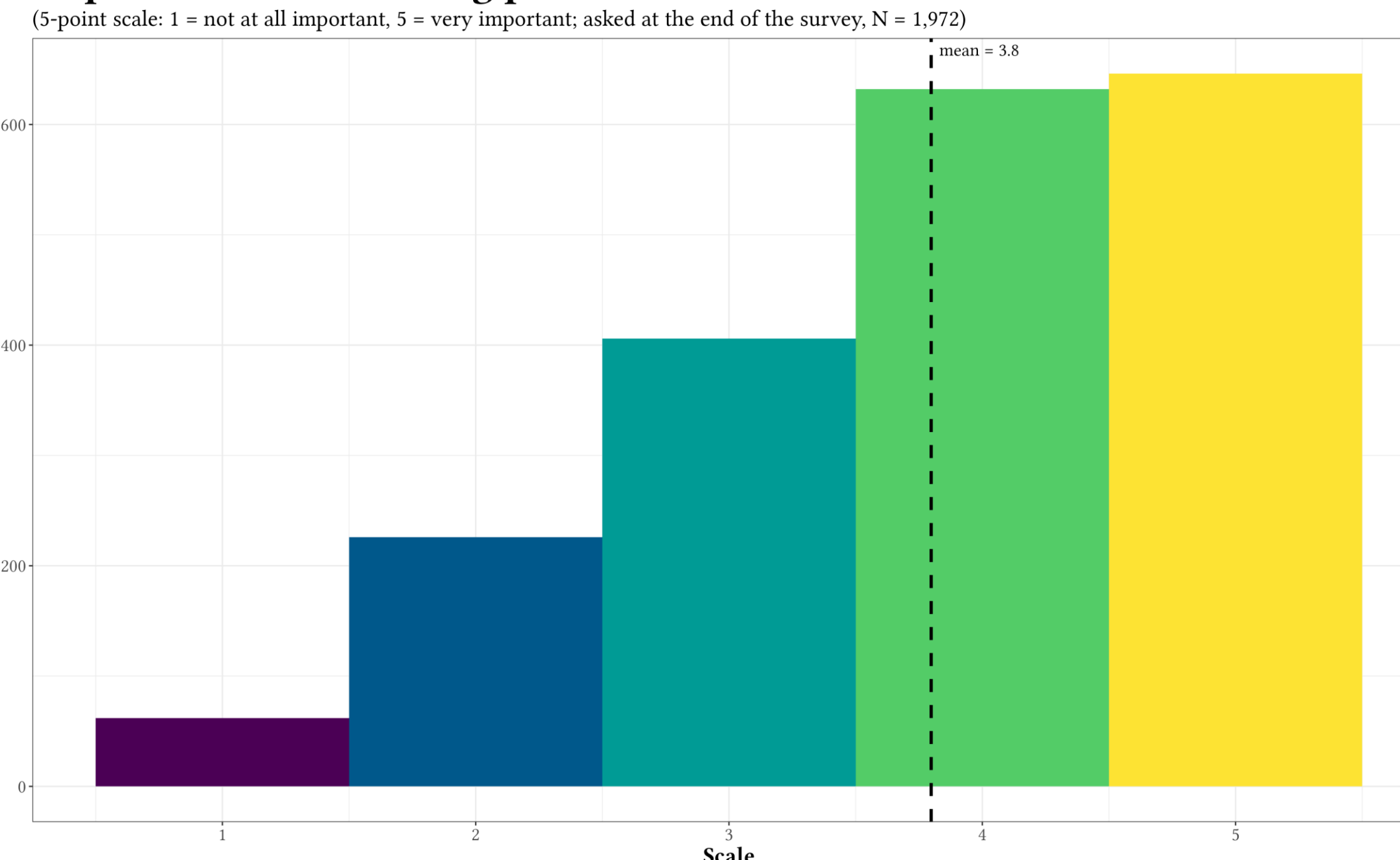
What Organization Should Benefit from XR Research with Children?



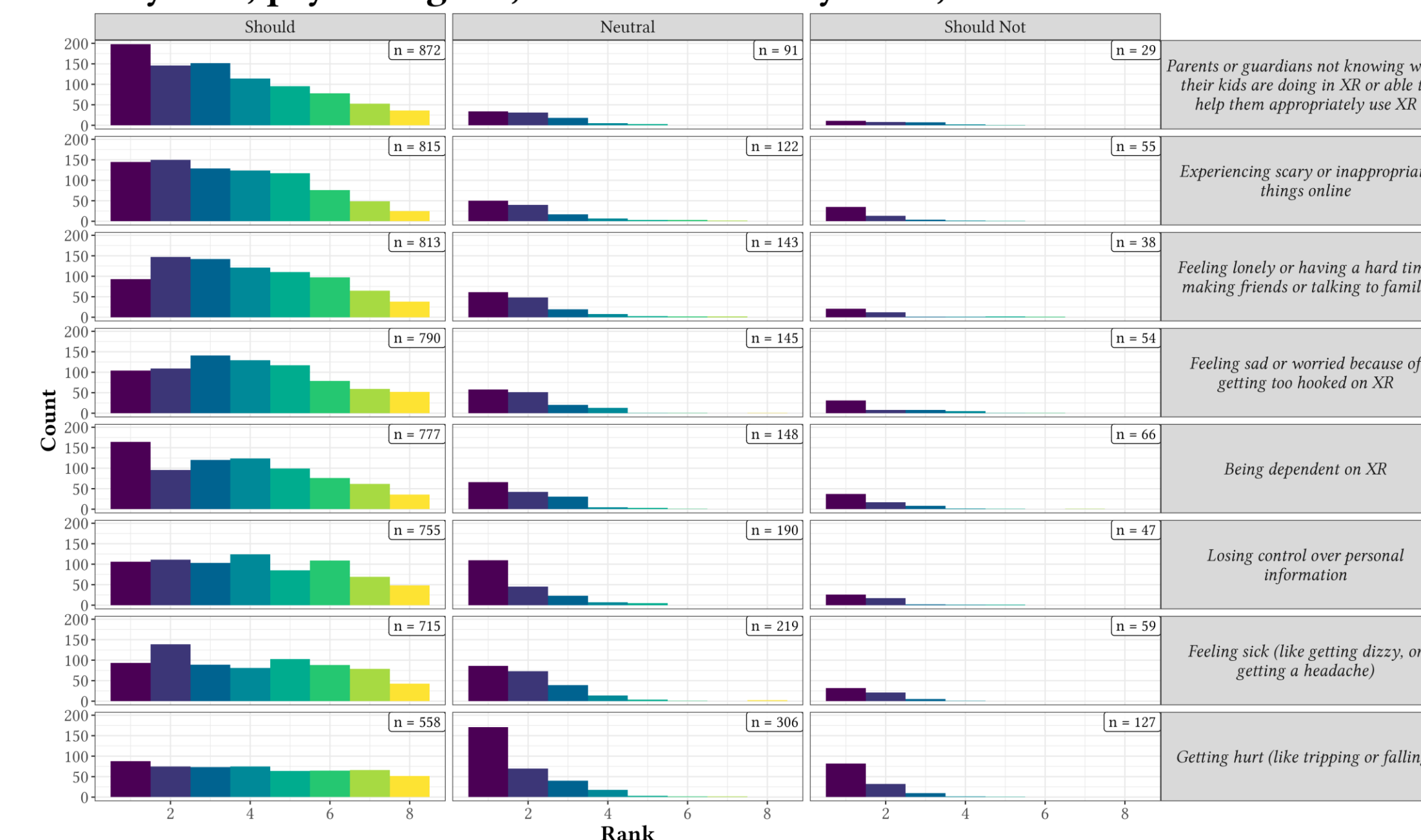
When Should Researchers Disclose when Data is Automatically Collected by XR Companies



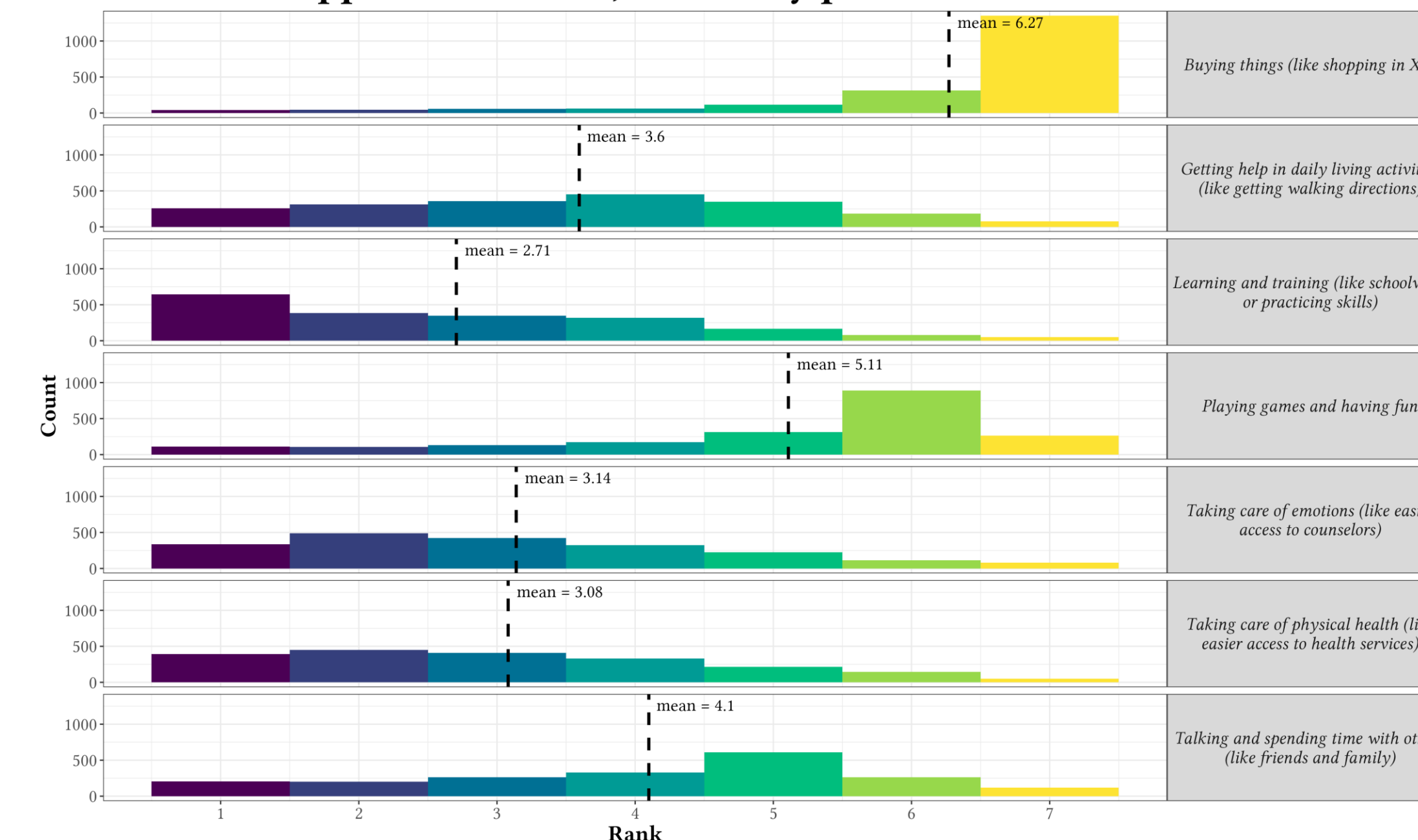
Importance of kids being part of XR research at all



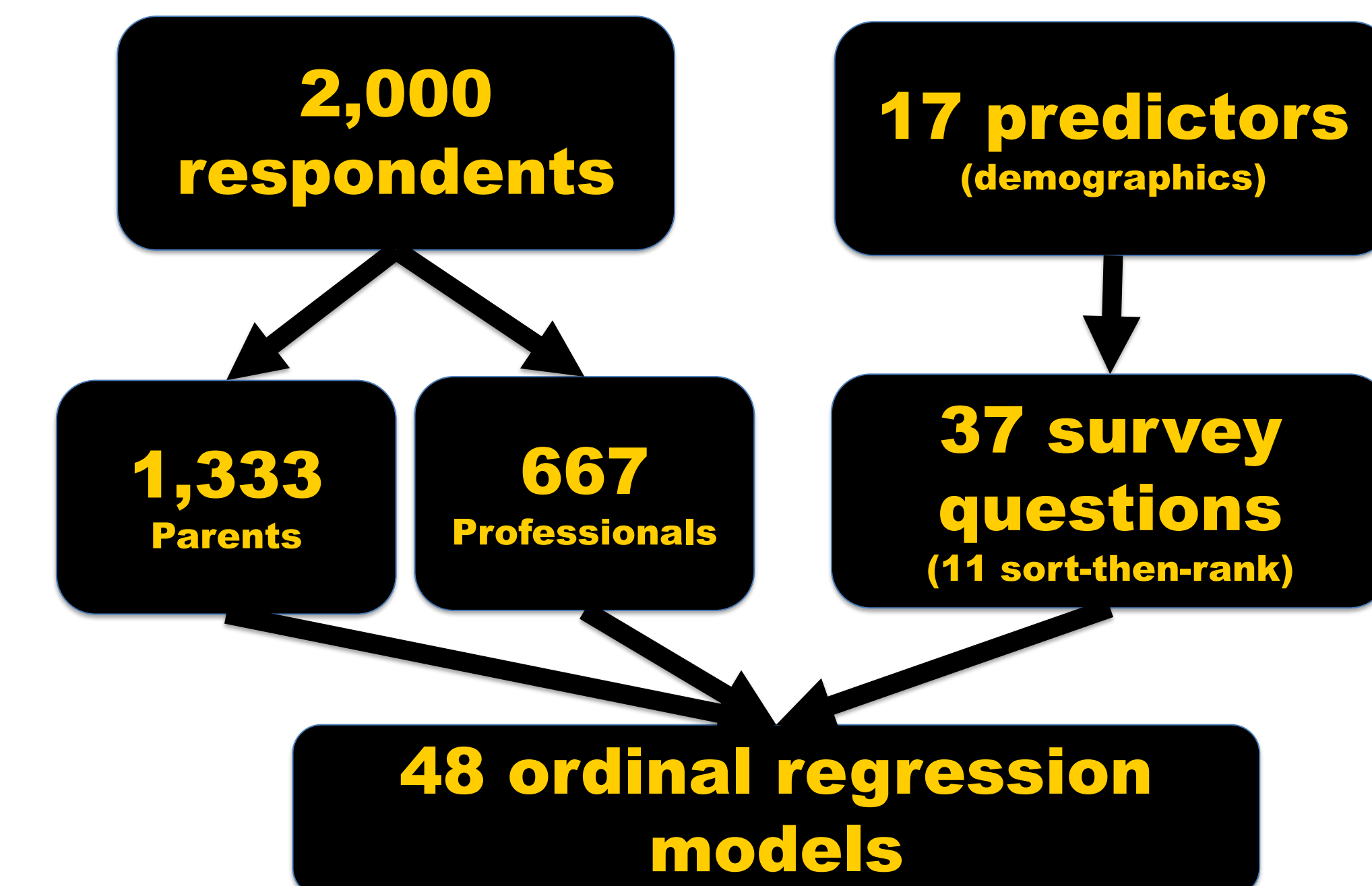
Physical, psychological, and online safety risks, ranked



Broad XR Application Areas, ranked by preference



Web Survey Setup



Findings

- Agreement on beneficial applications, but a safety research gap
- Mismatch in decision-making authority
- Consistent preference for inclusion of children with disabilities
- Individual variation in priorities and implications for stakeholder engagement

Limitations and Future Work

Respondents are not a representative sample of the entire United States' perspective, sharing the skewed demographic of those in professional positions working with children and parents. Children, a challenging population to include, were also excluded. Furthermore, during the implementation of the survey, respondents' individual race/ethnicity were not captured, instead opting to record census block data. Education level was missing for 551 individuals as well.

Future work includes creating more opportunities for other stakeholders in children's lives to give input, such as asking older siblings. Additionally, this survey will transition from nationwide to international to determine the broad ethical consensus.

Acknowledgements

I would like to give special thanks to Dr. Juan Pablo Hourcade for mentorship during this project and all my professors at the former Department of Statistics at the University of Nebraska-Lincoln who fought to uphold my right to a continued quality undergraduate education.

*Many results and demographic information were omitted. To see all tables and graphics, scan the QR code



References

- ³Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 1 (1995), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x
- ⁴Milton Friedman. 1937. The Use of Ranks to Avoid the Assumption of Normality Implicit in the Analysis of Variance. *J. Amer. Statist. Assoc.* 32, 200(1937), 675–701. doi:10.1080/01621459.1937.10503822
- ⁵Juan Pablo Hourcade, Meredith Onions, Amy Gilhoi, Delaney Norris, and Ethen Santana. 2026. Ethical Practices in XR Research with Children: A Systematic Review. doi:10.2139/ssrn.6929400
- ⁶Meta. 2026. Meta Quest Safety Information for Parents and Children. <https://www.meta.com/quest/parent-info/>
- ⁷Frank Wilcoxon. 1945. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* 1, 6 (1945), 80–83. doi:10.2307/3001968

